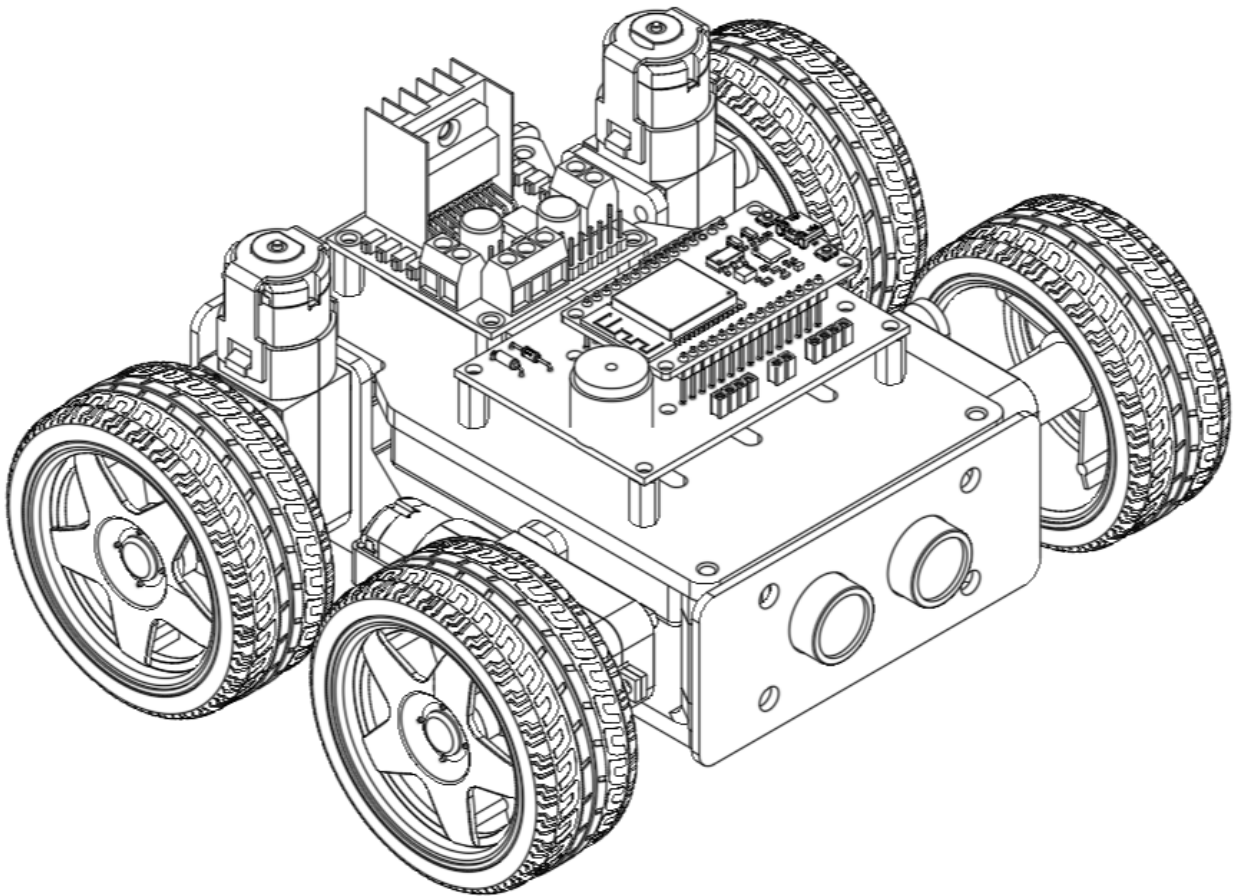


BUILD GUIDE

Robot Pathfinder



“Great discoveries and improvements invariably involve the cooperation of many minds.”

Alexander Graham Bell

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1. Introduction

This robot is intended for educational use only. It is named **Pathfinder**, after the Mars rover of the same name. Its overall dimensions are 180 × 150 × 90 mm and it incorporates numerous sensors dedicated to obstacle detection.

The robot can be used in three operating modes:

Mode	Description
Automatic	The robot is constantly alert and reacts to obstacles on its own: it detects obstacles, follows an object and can also detect drop-offs.
Semi-automatic (assisted driving)	The operator drives the robot, but the system enforces minimum distances from obstacles to keep the robot and bystanders safe.
Manual (free driving)	The operator keeps full control of the robot, with no restrictions.

Note Drop-off detection is disabled by default. To enable it, uncomment the corresponding section in the code.

2. Technical specifications

Characteristic	Specification
CAD software	Autodesk Fusion 360
Manufacturing process	3D printing (FDM)
Recommended material	Filament PETG (preferred): excellent mechanical strength and good impact durability
Overall dimensions	180 mm (length) × 150 mm (width) × 90 mm (height)

3. Bill of materials (BOM)

The table below lists all the components required to build the robot.

No.	Component	Reference / Description	Qty
1	Microcontroller	ESP32 Dev Module	1
2	Motor	DC gear motor	4
3	Motor driver	L293N	1
4	Ultrasonic sensor	HC-SR04	1
5	Infrared (IR) sensor	TCRT5000	2
6	White LED	5 mm, 5 V DC	4
7	Red LED	5 mm, 5 V DC	4
8	RGB LED	WS2812B	1
9	Resistor	220 Ω	4
10	Diode	1N4007	1
11	Female power jack	2.1 × 5.5 mm, round, off-center	1
12	Mini ON/OFF switch	KCD11, 3 A / 250 V, 2 or 3 pins	1
13	Lithium battery	18650	2
14	Electrical wire	—	As needed
15	Phillips or hex screw	M3-30	8
16	Phillips or hex screw	M3-8	16
17	Flat-head screw	M3-10	2
18	Female-female hex standoff	M3-30	4
19	Male-female hex standoff	M3-12	8
20	Female pin header	1 × 40 pins	2
21	Male pin header	1 × 40 pins	2

4. Assembly guide

Note All the parts that make up the robot's structure are available in the "Composent" folder. If the parts fit tightly during assembly, do not hesitate to enlarge the holes slightly.

Step 1: Mounting the motors

Solder wires of sufficient length onto each motor, then fasten them to the chassis with M3-30 screws, in the direction shown in the illustration.

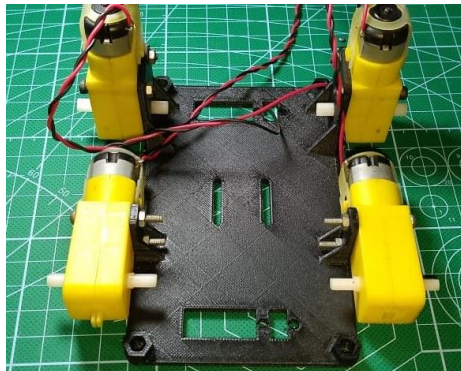


Figure 1 – Motors mounted on the chassis

Step 2: Infrared sensors and battery holder

Solder electrical wires onto the infrared (IR) sensors, then screw them onto the chassis together with the M3-30 hex standoffs. Then attach the battery holder with M3 flat-head screws.



Figure 2 – IR sensors and M3-30 standoffs



Figure 3 – Battery holder

Step 3: Fitting out the front and rear covers

Following the wiring diagram (section 5), solder the power jack, the switch and the four red LEDs, then install them in the rear cover. Next, solder the wires

to the ultrasonic sensor and the four white LEDs, then install them in the front cover.

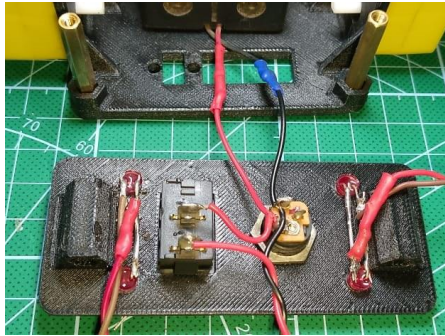


Figure 4 – Rear cover fitted

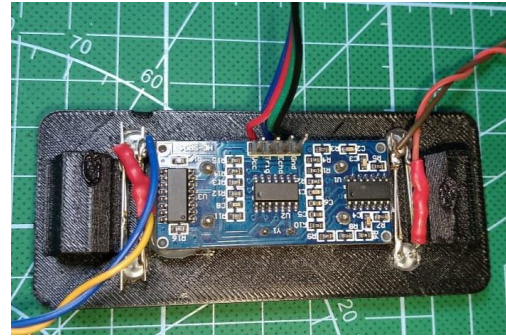


Figure 5 – Front cover fitted

Result At this point, the basic structure is mechanically ready and all front and rear peripherals are installed.



Figure 6 – Basic structure, front view



Figure 7 – Basic structure, rear view

Step 4: Closing the chassis

Place the roof on the chassis, making sure all the wires come out: the motor wires through the rear slots and the other cables through the front slots. Screw in the eight M3-12 standoffs and attach the motor driver module (see the bill of materials). Finally, insert all the M3 screws and tighten them to permanently join the roof to the chassis.

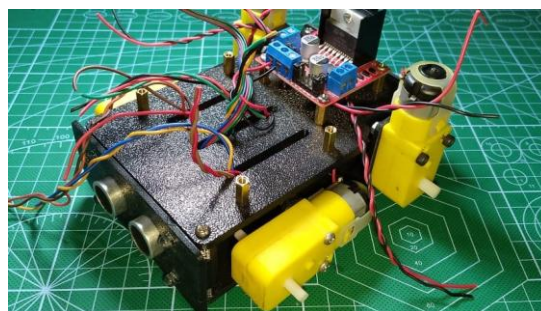


Figure 8 – Robot assembled with the roof and motor driver

5. Electronic circuit and main board

This section gathers all the diagrams and electrical connections of the robot.

5.1 Electronic circuit

Battery wiring

The diagram below shows the wiring of the battery (BT1), the power jack (J1) and the switch (SW1).

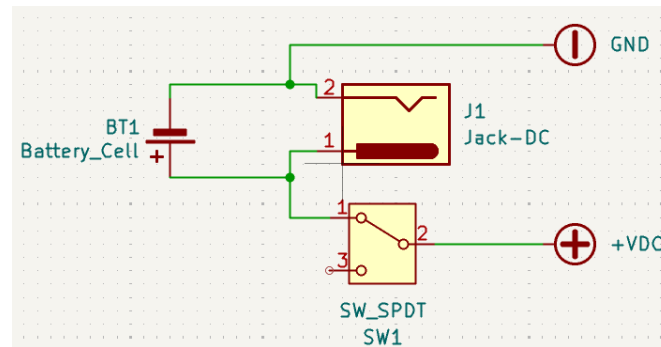


Figure 9 – Battery wiring diagram

LED wiring

The robot's lighting system is divided into four (04) independent groups. Each group consists of two (02) LEDs wired in parallel, the whole being placed in series with a 220 Ω resistor.

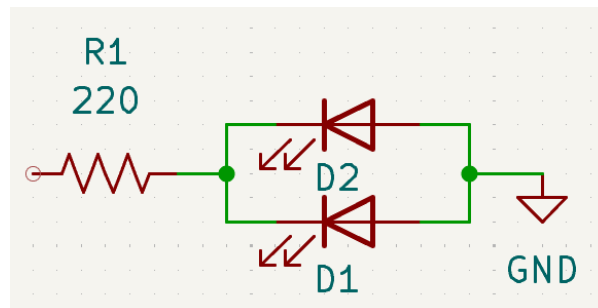


Figure 10 – Wiring diagram of one LED group

5.2 Main board

The main board is based on a 30-pin ESP32. All peripheral components are connected to it through male and female connectors carrying signals and power.

The board's main power is supplied by the +5 V output of the motor driver module.



Figure 11 – Main board installed on the chassis

5.2.1 Main board electronic schematic

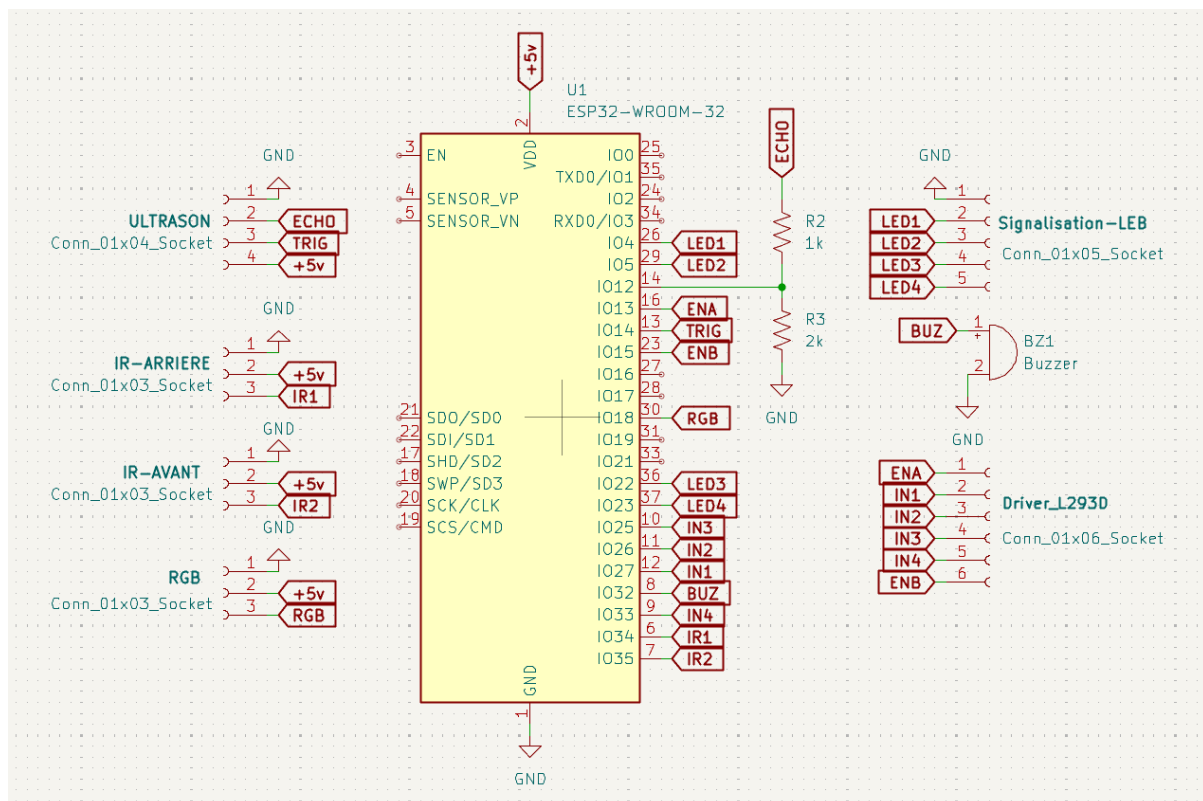


Figure 12 – Main board electronic schematic

5.2.2 ESP32-to-driver connection

Connect a 6-pin female connector to a 6-pin male connector to link the ESP32 and the motor driver.

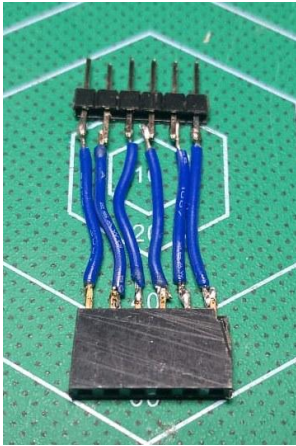


Figure 13 – 6-pin connector

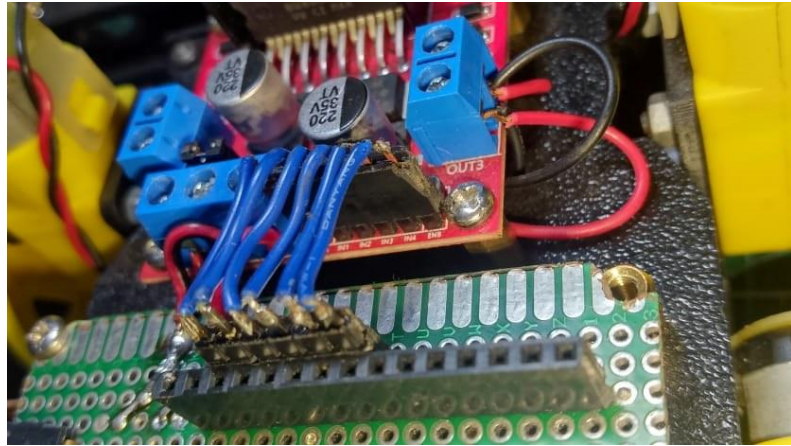


Figure 14 – Link to the motor driver

5.2.3 RGB LED connection

This is a WS2812B LED soldered onto a 3-pin male connector.



Figure 15 – WS2812B RGB LED module

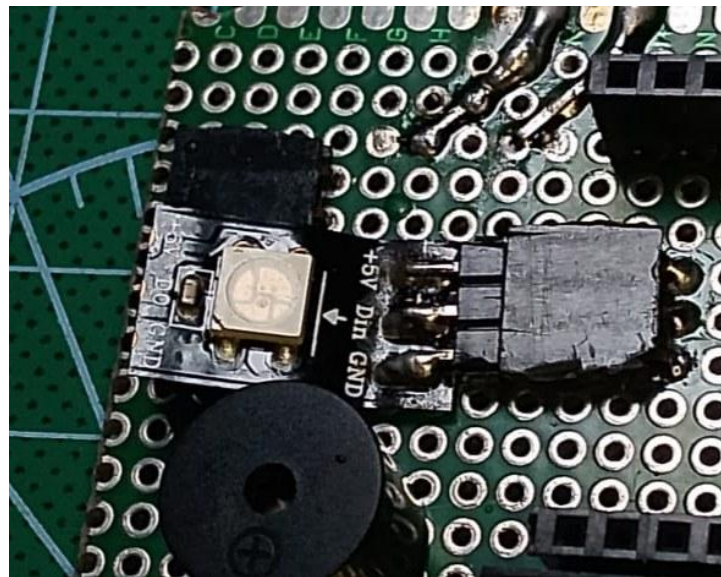


Figure 16 – Module installed on the main board

5.2.4 Signal LED connection

These are the LEDs built into the robot's front and rear covers. They are connected to the main board through male-female connectors, following the wiring diagram defined on the board.

To wire the LEDs to the connector:

- Connect all the LED grounds (GND) together, then make a common solder joint to the connector's ground pin.

- Then solder the LED control outputs to the adjacent connector pins, in the following order:
 1. Front cover, left LED
 2. Front cover, right LED
 3. Rear cover, left LED
 4. Rear cover, right LED

Important Respect the connection order and the LED polarity to ensure compatibility with the main board's electrical schematic.

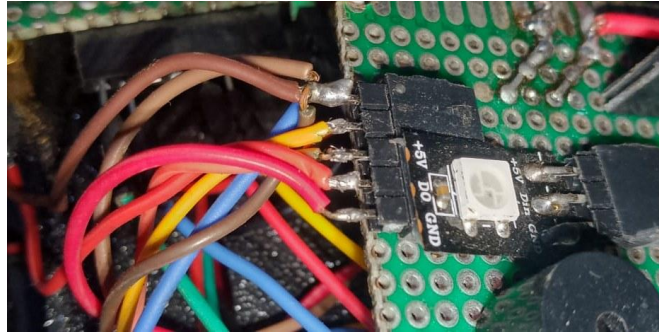


Figure 17 – Signal LED connections

5.3 IR sensor calibration

The infrared sensors must be calibrated to detect changes in distance between the robot and the surface it moves on. The goal is to identify any significant height difference indicating that part of the robot is over a void, or that the robot is abnormally raised.

Calibration consists of adjusting the potentiometer on each IR sensor to set the detection threshold correctly.

1. Power on the sensors and place the robot on a flat, uniform surface.
2. Observe the indicator LED on the IR module.
3. Slowly turn the potentiometer until the sensor reliably detects the surface beneath the robot.
4. Gradually move the sensor toward an area simulating a void (table edge, a black sheet held far away, or removing the surface under the sensor).
5. Adjust the potentiometer until the sensor's state changes exactly when the reference surface disappears.
6. Repeat several times, alternating the presence and absence of the surface, to check that detection is stable.
7. Make the same adjustment on the second IR sensor.

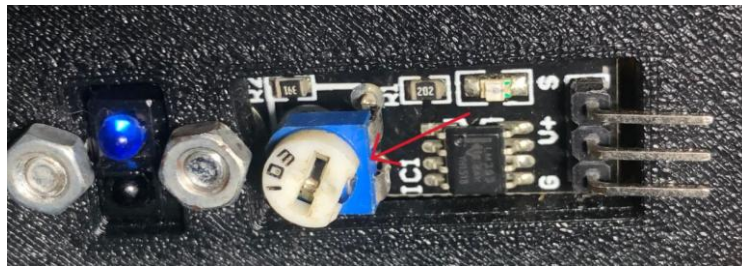


Figure 18 – TCRT5000 IR sensor and its adjustment potentiometer

6. Programs and source code

This section presents the programs required to run the robot. All files are available in the **Code source** folder.

6.1 Important tips

Uploading It is strongly recommended to remove the ESP32 board from its socket before uploading any code. Likewise, always switch the system off before putting it back.

Tests Run unit tests on each module to verify that it works correctly on its own.

Compilation Before compiling the code, install version 2.0.17 of the ESP32 package in the Boards Manager. If a newer version is installed, it is recommended to uninstall it to avoid compatibility conflicts.

6.2 Main programs

Program	Role
Auto-code	Manages the robot's full autonomy.
Mode-code	Smartphone control, with navigation between three sub-modes identified by the RGB LED color: Green: autonomous mode Blue: semi-autonomous mode Red: manual mode

7. App and control

The robot is controlled from the **Dabble** app, available on iOS and Android.

7.1 Connecting to the robot

After uploading the program to the board:

1. Open the Dabble app and go to the **Gamepad** section.
2. Connect to the device named **PathFinder**.



Figure 19 –
Gamepad
section in
Dabble

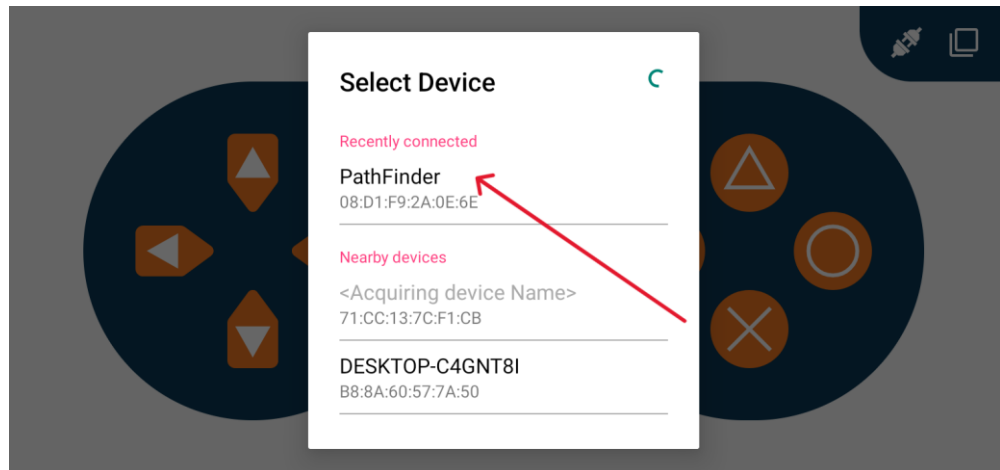


Figure 20 – Selecting the PathFinder device

7.2 Gamepad button functions

Button	Function
Arrow keys (UP, DOWN, LEFT, RIGHT)	Steer the robot.
SELECT	Select the robot's operating mode (manual, semi-automatic or automatic).
CROSS	Trigger the sound signal (buzzer).
TRIANGLE	Turn the front LEDs on and off.

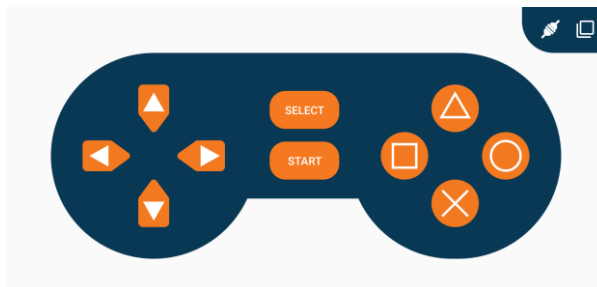


Figure 21 – Virtual gamepad of the Dabble app